

Supplementary Materials

Residual Limits of a Load-Normalized Spine-Neck Ratio in Compartmental Models

1 Extended Equations and Unit Notes

The implementation uses SI units internally at solver boundaries. Voltages are converted from mV to V, capacitances from pF to F, conductances from nS to S, currents from pA to A, and time from ms to s. Geometry-derived neck resistance uses centimeters for ρ_i compatibility:

$$R_{\text{neck}} = \frac{\rho_i L_n}{\pi r_n^2}. \quad (1)$$

For nonuniform neck profiles the code evaluates

$$R_{\text{neck}} = \int_0^{L_n} \frac{\rho_i}{\pi r(x)^2} dx \quad (2)$$

by numerical quadrature. Tapered, constricted, and profile-based necks preserve the intended geometry and report an effective resistance for comparison with lumped approximations.

The spine-omitted dendrite-soma load is a two-node DC circuit in the baseline reference case. With dendrite leak $g_{L,d}$, soma leak $g_{L,s}$, and dendrite-soma coupling g_{ds} , direct linear solution gives the dendritic one-port input resistance used in the main text:

$$R_{\text{in},d} = \left(g_{L,d} + \frac{g_{ds} g_{L,s}}{g_{ds} + g_{L,s}} \right)^{-1}. \quad (3)$$

Using the baseline values gives $R_{\text{in},d} = 144.486692 \text{ M}\Omega$, matching the Phase 4 direct DC solve to roundoff.

The analytic local divider is therefore

$$\Gamma_{\text{div}} = \frac{R_{\text{in},d}}{R_{\text{neck}} + R_{\text{in},d}} = \frac{1}{1 + \text{SMI}}, \quad (4)$$

and the manuscript residual is

$$\Delta_{\text{div}} = \Gamma_{h \rightarrow d} - \Gamma_{\text{div}}. \quad (5)$$

These equations define a DC small-signal reference. They do not assert that transient conductance-based peak transfer must equal the divider value.

The passive matrix in the main manuscript generalizes to an arbitrary passive tree as

$$\mathbf{C} \frac{d\mathbf{V}}{dt} = -(\mathbf{G}_L + \mathbf{G}_A + \mathbf{G}_{\text{syn}}(t)) \mathbf{V} + \mathbf{b}(t), \quad (6)$$

where $\mathbf{G}_A$ is assembled from axial edges. Each edge with conductance g_{ij} contributes $+g_{ij}$ to both diagonal entries and $-g_{ij}$ to the symmetric off-diagonal entries. This construction preserves reciprocity in passive linear systems.

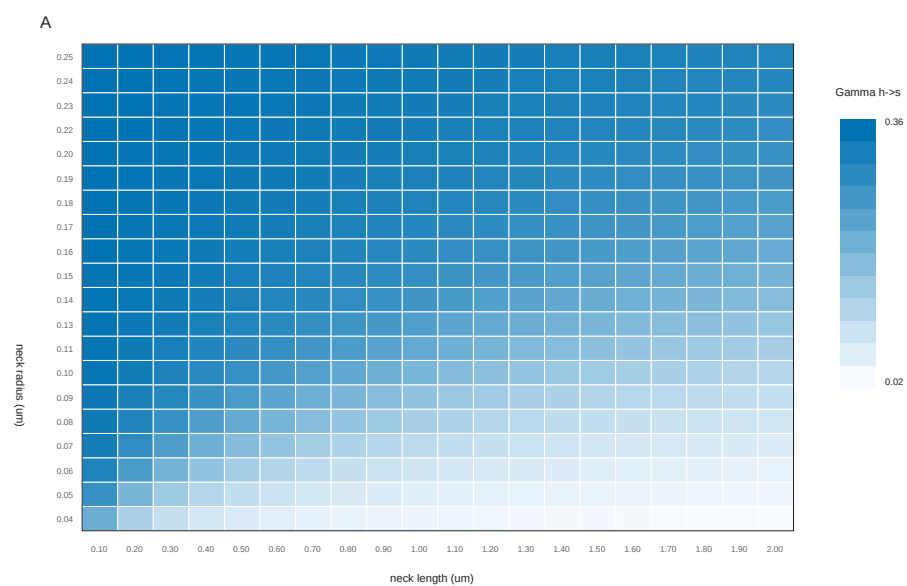


Figure 1: Figure S1. Fixed-load head-to-soma transfer heatmap from the baseline validation sweep. The heatmap complements the main-text load-normalization controls by showing the two-parameter neck-geometry grid.

2 Numerical Validation

The passive baseline was checked against geometry identities, synaptic normalization, current signs, matrix limits, timestep refinement, high-synapse stability, and Backward Euler versus Crank-Nicolson agreement. The baseline validation script generated the passive source data and classified all representative reference target values as passing predefined target-case tolerances. The most stringent target-case tolerance class means absolute error ≤ 0.005 or relative error $\leq 2\%$, not bitwise equality.

Phase 4 added an independent direct-matrix benchmark for the passive three-compartment baseline reference cases. The benchmark independently parsed the baseline parameters, recomputed capacitance/leak/neck terms and the double-exponential synaptic conductance, assembled the passive matrices, and solved the Backward Euler system without calling the production passive simulation function as its benchmark engine. Compared with existing SPINE trace CSVs, the maximum all-trace absolute difference was 1.25×10^{-13} mV, the maximum head-amplitude difference was 1.42×10^{-14} mV, and the maximum $\Gamma_{h \rightarrow d}$ difference was 7.55×10^{-15} .

Phase 4 also checked DC analytic limits. The spine-omitted two-node load formula gave $R_{in,d} = 144.48669201520912$ M Ω , and the direct DC solve gave 144.48669201520914 M Ω . Divider predictions for the low, intermediate, and high reference cases were 0.9912648, 0.8970688, and 0.4306959. The zero-neck and very-large-neck limiting cases approached local transfer values of 1 and 0, respectively.

The BE-versus-CN reference-case comparison remains an internal scheme-consistency check. Across the three baseline reference cases, the maximum peak differences were 0.00203 mV for A_h , 0.000534 mV for A_d , 0.000189 mV for A_s , 0.000500 for $\Gamma_{h \rightarrow d}$, and 0.000180 for $\Gamma_{h \rightarrow s}$. The maximum trace voltage difference was 0.112 mV in the high-SMI case.

Passive morphology validation included sparse versus dense equivalence, spatial convergence, sinusoidal impedance validation, logarithmic-chirp impedance validation, and distributed-neck comparisons. Sinusoidal validation produced maximum relative amplitude error 0.0054. Chirp validation produced maximum relative amplitude error 0.072. Spatial convergence passed at the 16-versus-32 segment comparison with relative $\Gamma_{h \rightarrow s}$ difference 0.014.

Active validation included AMPA peak normalization, NMDA magnesium-block monotonicity, NMDA current-voltage relief, active-current sign checks, gate bounds, positive gate time constants, resting-state stability, timestep refinement, limiting-case solver comparison, strong-drive stability, and active unit checks. All 12 active validation rows passed.

NEURON was unavailable in the Phase 4 execution environment. No NEURON result is claimed, and NEURON is not a runtime dependency of SPINE.

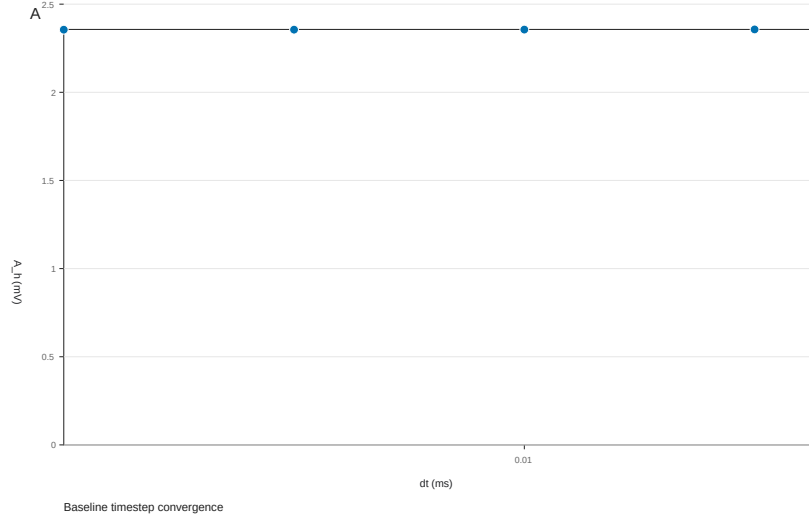


Figure 2: Figure S2. Timestep convergence for the baseline-reference intermediate condition. The panel reports the audited head-amplitude convergence check used for numerical validation.

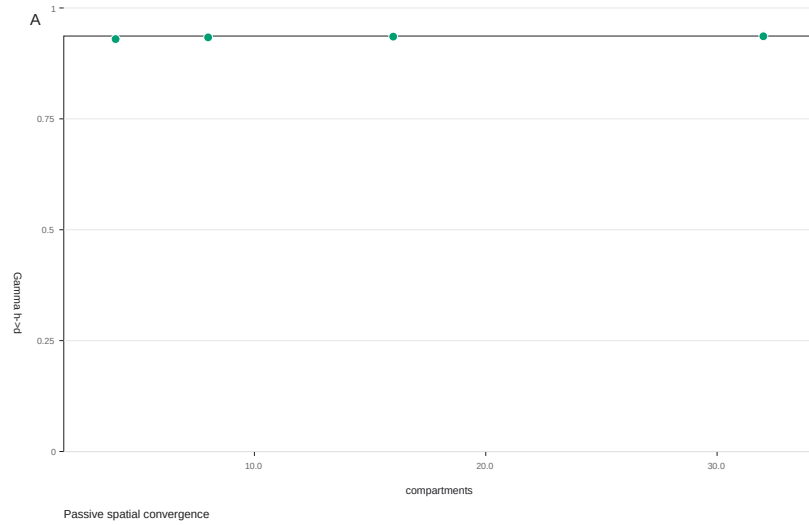


Figure 3: Figure S3. Spatial convergence for the passive morphology engine. The panel shows the dependence of local transfer on compartment discretization.

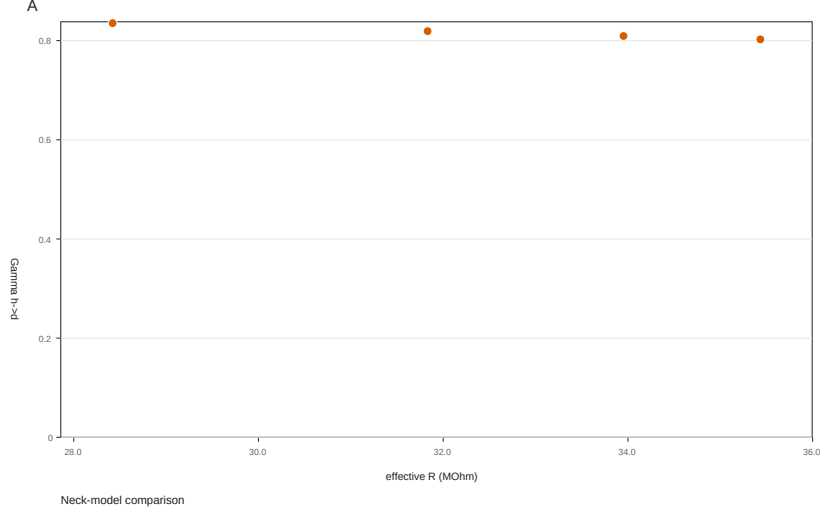


Figure 4: Figure S4. Lumped and distributed neck-model comparisons. The panel contrasts effective neck resistance with local transfer for audited passive neck variants.

3 Uncertainty, Identifiability, and Predictor Stability

The uncertainty analysis used deterministic Latin-hypercube ensembles and deterministic resampling diagnostics. A robustness audit extended this to progressive $N=96, 192, 384$, and 768 ensembles. The global uncertainty seed was 202605 ; progressive-sampling ensembles used deterministic offsets $202605 + 5100 + N$. The original 48-versus-96 SMI convergence failure was preserved and interpreted as a small-sample warning in a skewed SMI distribution. By $N=768$, final SMI median convergence passed with relative change 0.012 , and final all-median convergence passed with maximum relative change 0.029 .

Radius uncertainty was analyzed independently because of the quadratic radius term in R_{neck} . The radius audit used 25 deterministic offsets spanning ± 2 SD, with $SD = 0.018 \mu\text{m}$ and clipping to $0.06\text{--}0.22 \mu\text{m}$. At $N=768$, 184 of 768 sampled parameter combinations changed SMI class under the predefined radius-error model, a design fraction of 24.0% . Low-class combinations flipped in 22.9% of cases, all 11 intermediate combinations flipped, and no high-SMI combinations were present. The median radius-induced SMI fold range was 3.44 . These values are deterministic design summaries, not biological rates.

High-SMI uncertainty coverage is absent in the main $N=768$ deterministic ensemble. The ensemble contains 757 low-SMI combinations, 11 intermediate combinations, and zero high-SMI combinations. High-SMI behavior in the manuscript is therefore supported by reference and designed challenge cases rather than by the $N=768$ uncertainty ensemble.

Predictor comparisons used Pearson association, Spearman association, deterministic resampling summaries, cross-validated RMSE, and multivariable diagnostics. Exact top predictor labels were not fully stable because near-tied dynamic-SMI variants changed order. The manuscript therefore emphasizes descriptor families and mechanistic interpretation.

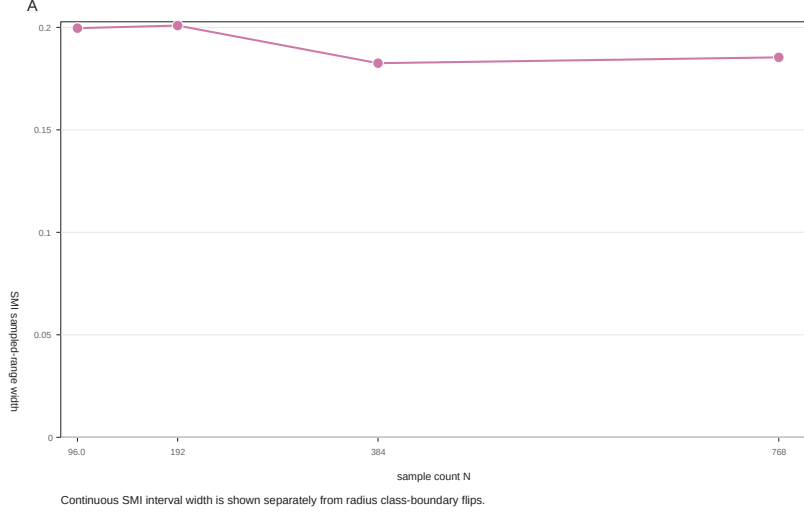


Figure 5: Figure S5. Continuous SMI uncertainty-width convergence across progressive sample counts. This continuous-width check complements the main-text radius class-flip design fraction.

4 Active and Nonlinear Mechanism Notes

AMPA and NMDA synapses are opt-in extensions. NMDA includes a voltage-dependent magnesium block based on established reduced forms [1, 2]. Na and delayed-rectifier K conductances follow Hodgkin-Huxley-style conductance-current logic [3]. HCN, A-type K, and restrained calcium conductances are included as generic active-state stressors with provenance from dendritic physiology literature [4–7].

These conductances are not cell-type-calibrated densities. They are disabled in the baseline reference configuration and should not be interpreted as fitted biological parameters. Their purpose in this manuscript is to test whether passive load-ratio conclusions survive nonlinear and active stress tests.

Table 1: Active-extension parameter audit. All mechanisms are disabled in the baseline reference configuration and enabled only through active-extension configurations or explicit protocols.

Mechanism	Main parameters	Values	Units	Provenance/status
AMPA	$g_{\max}$; τ_r ; τ_d ; E_{rev}	1.4; 0.3; 3.0; 0	nS; ms; ms; mV	active-extension double exponential
NMDA	$g_{\max}$; τ_r ; τ_d ; Mg ; η ; γ	0.7; 2.0; 80.0; 1.0; 3.57; 0.062	nS; ms; ms; mM; mM; mV^{-1}	reduced magnesium block family
Na	$\bar{g}$; E_{rev} ; gates	6.0; 55; m^3h	nS; mV; dimensionless	restrained HH-style extension
KDR	$\bar{g}$; E_{rev} ; gates	1.5; -90; n^4	nS; mV; dimensionless	restrained HH-style extension
HCN	$\bar{g}$; E_{rev} ; gates	0.05; -30; r	nS; mV; dimensionless	generic subthreshold HCN approximation
A-type K	$\bar{g}$; E_{rev} ; gates	0.5; -90; a^3b	nS; mV; dimensionless	restrained transient potassium extension
CaT	$\bar{g}$; E_{rev} ; gates	0.05; 120; p^2q	nS; mV; dimensionless	electrical-only restrained calcium extension

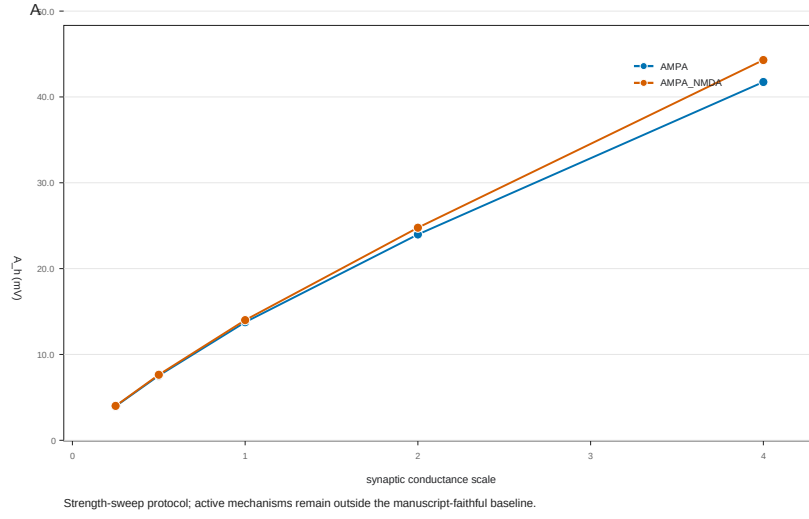


Figure 6: Figure S6. Active AMPA and AMPA+NMDA strength-sweep protocol. The panel shows head-amplitude responses across pre-specified conductance scales for the active-extension protocol library.

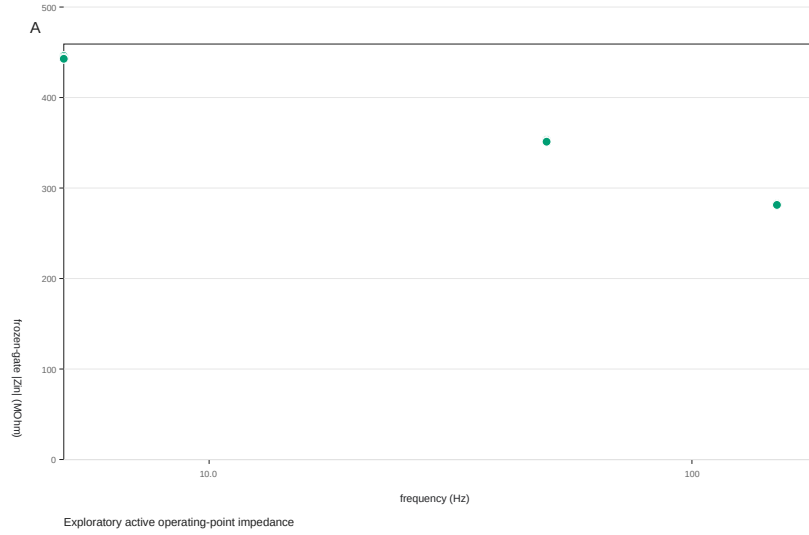


Figure 7: Figure S7. Exploratory frozen-gate operating-point impedance. The panel summarizes operating-point impedance estimates used as exploratory active-context descriptors.

5 Exploratory Perturbation Scenarios

The separated exploratory module uses restrained perturbation scenarios involving morphology, HCN, potassium, AMPA/NMDA, clustering, synchrony, combined perturbations, and conflicting directions [8–12]. The source rows explicitly mark these scenarios as exploratory-only and make no diagnostic or therapeutic claim.

The exploratory uncertainty screen uses $N=8$ sampled combinations per scenario. It is a sensitivity screen rather than a biological uncertainty estimate or rare-tail analysis. The exploratory scenarios are best cited as examples showing that synaptic strength, active state, clustering, and timing can alter voltage outcomes without changing morphology-defined SMI.

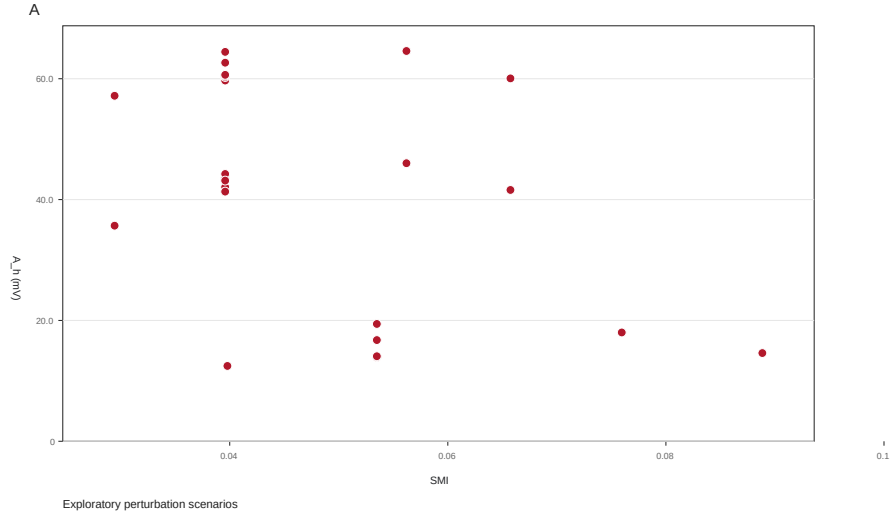


Figure 8: Figure S8. Exploratory perturbation scenarios showing SMI versus head amplitude. These cases remain separated from the general model and are not used to support primary claims.

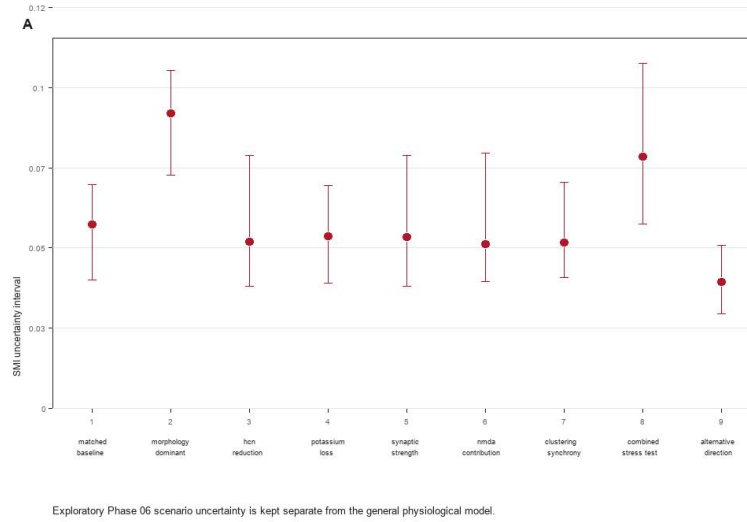


Figure 9: Figure S9. Scenario-level SMI uncertainty screen. Intervals summarize exploratory perturbation scenarios and are presented as context rather than primary validation evidence.

6 Reproducibility Commands

The following commands reproduce the validated software checks and manuscript source-data generation in a local development environment. Broad primary ensembles should not be rerun casually during review unless the authors intentionally regenerate the source data.

Windows PowerShell:

```
$env:PYTHONPATH='src'
python -m unittest discover -s tests -p 'test_*.py'
python -m spine.cli smoke configs\manuscript_faithful\baseline.toml '
--output-csv results\phase01_smoke_trace.csv
python scripts\reproduce_manuscript.py
python scripts\revision_restart\phase4_validation_runner.py
python scripts\revision_restart\phase7_generate_divider_residual_figure.py
```

Unix shell:

```
PYTHONPATH=src python -m unittest discover -s tests -p 'test_*.py'
PYTHONPATH=src python -m spine.cli smoke configs/manuscript_faithful/baseline.toml \
--output-csv results/phase01_smoke_trace.csv
PYTHONPATH=src python scripts/reproduce_manuscript.py
PYTHONPATH=src python scripts/revision_restart/phase4_validation_runner.py
PYTHONPATH=src python scripts/revision_restart/phase7_generate_divider_residual_figure.py
```

Reviewer-access package drafts include manifests and checksums. Private access, public release, archival deposition, and license selection remain subject to author, PI, institutional, and journal approval.

7 Extended Limitations

SPINE is a transparent computational framework, not a calibrated biological reconstruction. The baseline reference track preserves pre-specified assumptions even when they are simplified. The

plausibility-revised and active-extension tracks are explicitly separated. Exploratory perturbation scenarios are supplemental stress tests and are not disease models.

The analytic divider is a DC small-signal reference. The simulated targets in this manuscript are transient conductance-based peak responses, so residuals from the divider are expected and informative. The independent direct-matrix benchmark checks the passive three-compartment baseline implementation against an independently assembled equation solve, but it does not check every morphology, active, uncertainty, or exploratory output. NEURON cross-checking was not performed.

The deterministic uncertainty ensembles are sensitivity designs, not experimentally fitted posterior distributions. The N=768 ensemble contains no high-SMI rows, so high-isolation uncertainty remains a future targeted extension. Radius uncertainty, unmodeled ultrastructure, cytoplasmic resistivity uncertainty, segmentation error, local synaptic receptor composition, and active conductance distributions can all alter interpretation.

The current package does not include posterior inference from experimental voltage traces, cell-type-specific channel fitting, full morphology libraries, biochemical calcium reaction-diffusion, receptor trafficking, signaling cascades, or public release approvals. These are future validation and administrative targets, not hidden assumptions of the current manuscript.

References

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